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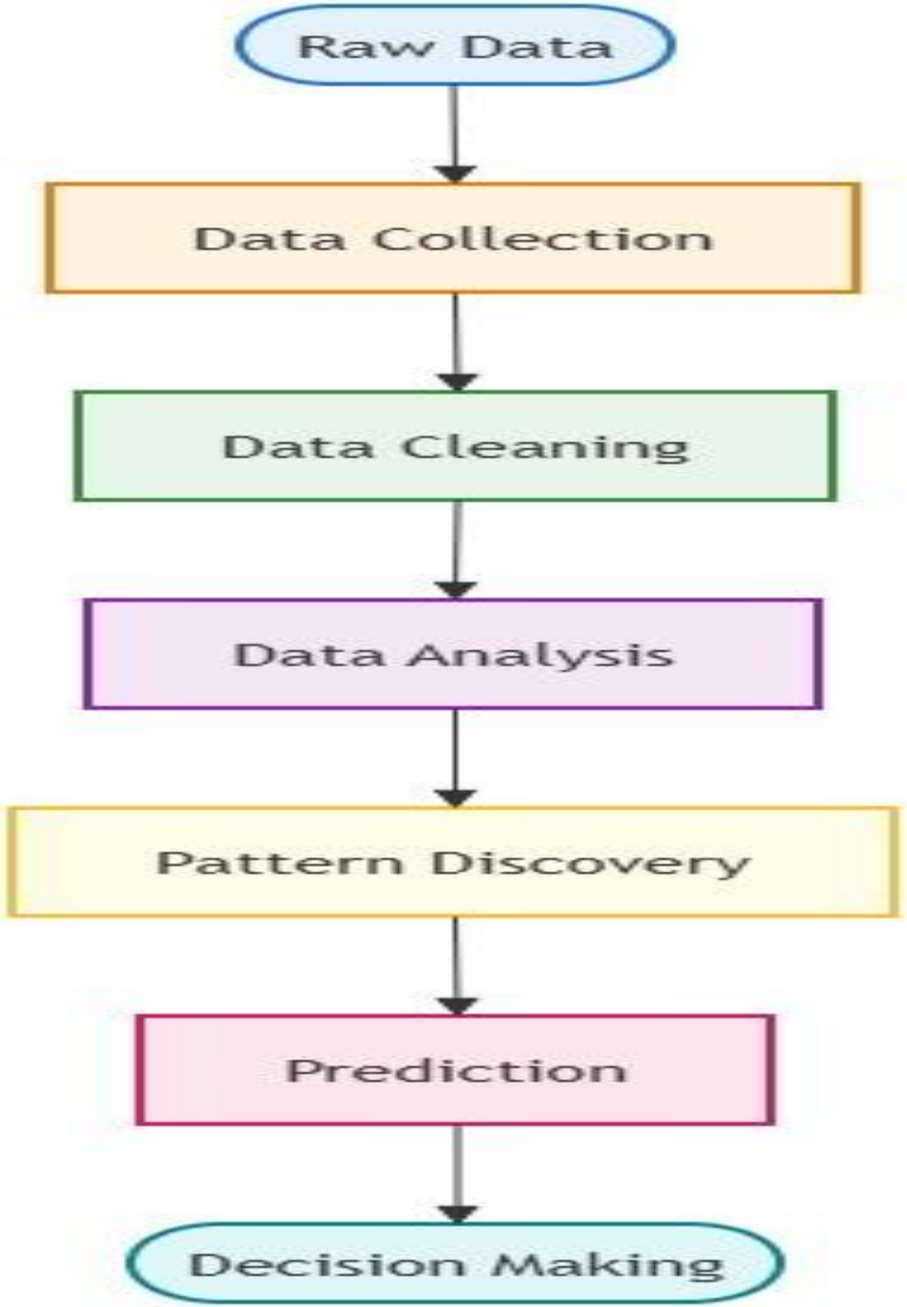
UNIT 1

Introduction to Data Science & Python for Data Science

Learning Objective

- Explain Big Data and its characteristics
- Roles and Responsibilities of a Data Scientist
- Data Scientist's Toolbox Data Science
- Challenges Exploratory Data Analysis (EDA)
- Python for Data Analysis
- Data Science Editors Using Python

DATA SCIENCE



Data Science is an interdisciplinary field that collects, cleans, analyzes, and interprets data to discover meaningful insights and support better decision-making

Data Science is the process of collecting, cleaning, analyzing, and interpreting data to solve real-world problems and support decision-making.

What is Big Data?

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- Big Data refers to extremely large and complex datasets that cannot be processed efficiently using traditional data processing tools.
- Big Data is a collection of very large amounts of data that is generated every second and requires advanced tools to store, process, and analyze it.

Example: Every minute, millions of users:

- Upload videos
- Watch videos
- Like and comment
- Search for content

This generates terabytes of data every day, YouTube data is an example of Big Data.

Example

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Amazon

Every second, customers:

- Search for products
- Add items to their cart
- Make purchases
- Write reviews

Amazon analyzes this Big Data to recommend products such as **"Customers who bought this also bought..."**

Healthcare

Hospitals generate data from:

- Patient records
- Blood test reports
- X-rays and MRI scans
- Wearable devices (smartwatches)

Doctors use this data to predict diseases and improve treatment.

The 4 Vs of Big Data

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V	Meaning	Unit / Measurement	Real-Life Example	Technology Used
Volume	Amount or size of data generated	**KB, MB, GB, TB, PB, EB, ZB**	YouTube stores petabytes of videos every day	Hadoop HDFS, Cloud Storage
Velocity	Speed at which data is generated and processed (Refers to how **quickly data is generated, collected, and processed**.)	**Records/second, Transactions/second (TPS), MB/s, GB/s**	Credit card transactions, live stock market updates	Apache Kafka, Apache Spark
Variety	Different types and formats of data	**Structured, Semi-structured, Unstructured**	Text, Images, Videos, Audio, Emails, Sensor data	NoSQL Databases (MongoDB, Cassandra)
Veracity	Quality, accuracy, and reliability of data (Refers to the **reliability and correctness** of data. Poor-quality data can lead to incorrect analysis.)	**Accuracy (%), Error Rate (%), Data Quality Score**	Removing duplicate records, correcting missing or incorrect values	Data Cleaning, ETL Tools, Python Pandas

Example

Suppose you use YouTube every day. Every second, thousands of people upload videos, millions watch them, write comments, and press Like. YouTube must store all these videos (Volume), process them instantly (Velocity), handle different data such as videos, text, and images (Variety), and ensure the information is accurate by removing spam or fake content (Veracity). These four characteristics together are called the 4 V's of Big Data.

Who is a Data Scientist?

A Data Scientist is a professional who collects, analyzes, and interprets data to solve business problems and support decision-making.



Skills

- Programming
- Statistics
- Machine Learning
- Communication
- Domain Knowledge

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Roles & Responsibilities

Responsibility	Description
Business Understanding	Identify the business problem
Data Collection	Gather data from multiple sources
Data Cleaning	Remove errors and missing values
Data Analysis	Discover patterns
Model Building	Develop ML models
Model Evaluation	Measure performance
Deployment	Implement the model
Communication	Present insights using reports and dashboards

Data Scientist's Toolbox

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Programming

- Python
- R
- SQL

Libraries

- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn

Visualization

- Tableau
- Power BI

Development

- Jupyter Notebook
- Google Colab
- VS Code











Data Science Challenges

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Common Challenges

- Poor Data Quality
- Missing Values
- Duplicate Records
- Large Data Volume
- Privacy & Security
- Biased Data
- High Computational Cost

Best Practices

✓ Validate data before analysis

✓ Document all assumptions clearly

✓ Test and evaluate models properly

✓ Choose visuals suited to the data

Example

In the Pima Diabetes Dataset, values such as Insulin = 0 are not realistic and are treated as missing values before analysis.

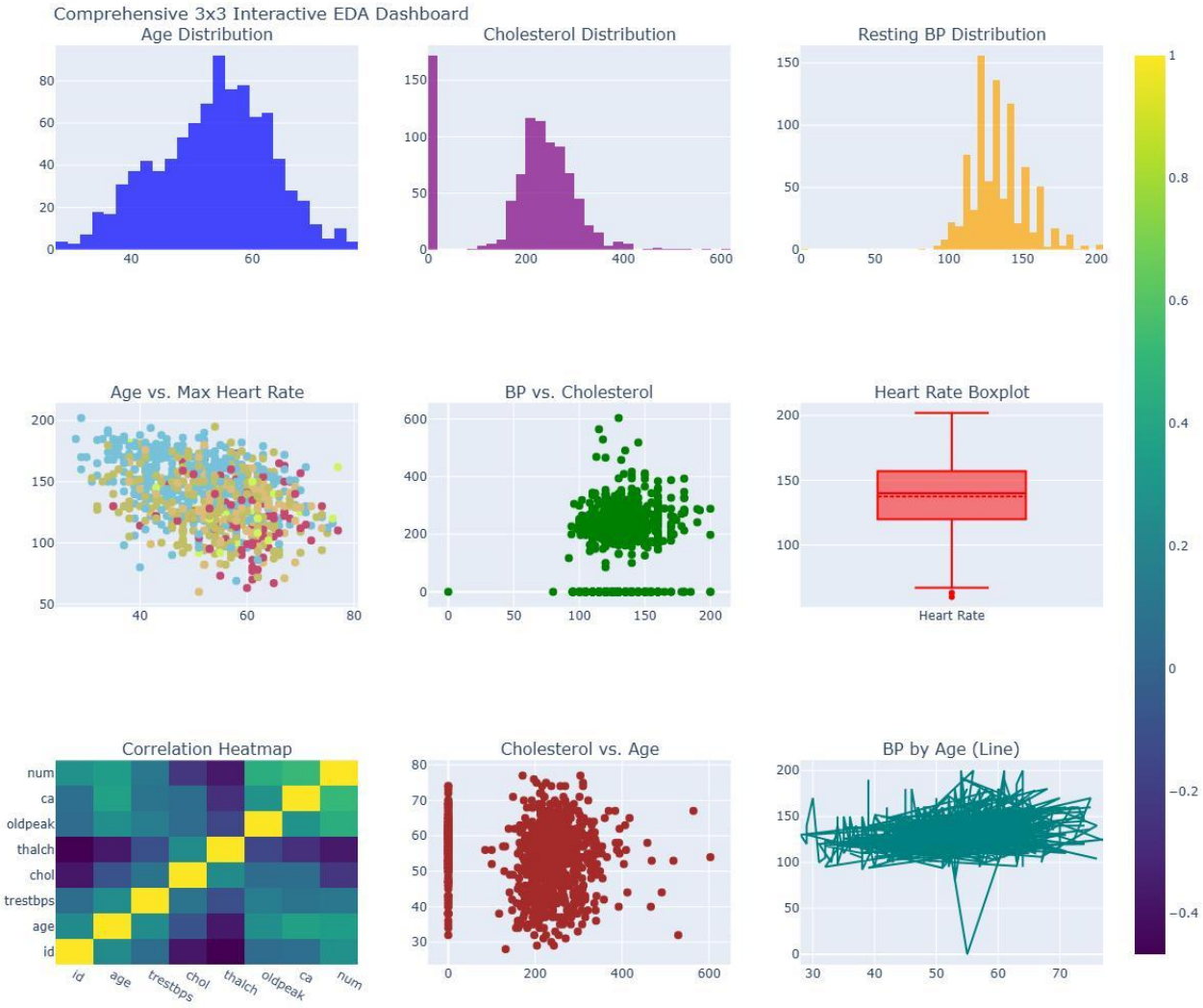
Exploratory Data Analysis (EDA)

EDA is the process of exploring and understanding a dataset before building a Machine Learning model.

Objectives

- Understand the dataset
- Find missing values
- Detect outliers
- Study relationships
- Visualize patterns

Load Dataset
↓
Understand Dataset
↓
Check Missing Values
↓
Summary Statistics
↓
Visualize Data
↓
Find Patterns



Exploratory Data Analysis (EDA) Operations with Python

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1. Load the Dataset:Read the CSV file into a DataFrame.

```
import pandas as pd
df = pd.read_csv("diabetes.csv")
```

2. Display First 5 Records: View the first five rows of the dataset.

```
df.head()
```

3. Display Last 5 Records**:** View the last five rows.

```
df.tail()
```

4. Check Dataset Shape :Find the number of rows and columns.

```
df.shape
```

5. Display Column Names:List all feature names.

```
df.columns
```

6. Check Dataset Information: Display data types, non-null values, and memory usage.

```
df.info()
```

7. Generate Statistical Summary :Display statistical information (mean, min, max, etc.).

```
df.describe()
```

8. Check Missing Values : Count missing values in each column.

```
df.isnull().sum()
```

9. Replace Zero Values with NaN : Replace invalid zero values with missing values.

```
import numpy as np
cols =["Glucose","BloodPressure","SkinThickness","Insulin","BMI"]
df[cols] = df[cols].replace(0, np.nan)
```

10. Fill Missing Values : Replace missing values with the median.

```
for col in cols:
    df[col] = df[col].fillna(df[col].median())
```

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Why Python for Data Science

Why Python?

Python is the most popular programming language in Data Science because it is simple, readable, and supported by a rich ecosystem of libraries — from data wrangling to deep learning.

DEFINITION

Python lets a data scientist move from raw data to a working model in the same language, without switching tools for statistics, visualization, or deployment.

AT A GLANCE

- Beginner-friendly, readable syntax
- Free and open source
- Huge ecosystem: NumPy, Pandas, Scikit-learn...
- Runs everywhere: Windows, Linux, macOS

Why Python?

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Python vs. Other Languages

Language	Strength	Common Use in Data Science
Python	General-purpose, huge DS library ecosystem	End-to-end: cleaning, modeling, deployment
R	Deep statistical & academic package support	Statistical research, specialized visualization
Java / Scala	High performance on large distributed systems	Big-data pipelines (e.g. Apache Spark)
SQL	Declarative querying of relational data	Extracting and aggregating stored data

Python's advantage is breadth — it covers the whole workflow in one language, from raw data to a deployed model.

IPython Shell

IPython (Interactive Python) is an enhanced interactive Python shell that provides a more powerful environment than the standard Python interpreter.

FEATURES

- Interactive coding
- Syntax highlighting
- Auto-completion (Tab key)
- Command history
- Magic commands (%time, %run)
- Easy debugging

```
In [1]: import numpy as np
In [2]: arr = np.array([1,2,3])
In [3]: arr.mean()
Out[3]: 2.0

In [4]: %time arr.sum()
```

Advantages: faster experimentation · better error messages · convenient for testing small code snippets

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Large Collection of Libraries

Library	Purpose
NumPy	Numerical computations
Pandas	Data analysis and manipulation
Matplotlib	Data visualization
Seaborn	Statistical visualization
Scikit-learn	Machine Learning algorithms
TensorFlow	Deep Learning
Keras	Neural Networks

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Data Science Editors Using Python

Jupyter Notebook

Jupyter Notebook is an interactive web-based environment used to write Python code, visualize data, and document analysis in one place.

Features

- Interactive coding
- Markdown support
- Data visualization
- Easy sharing
- Automatic saving

Jupyter Notebook

- Interactive environment for writing Python code.
- Displays code, output, graphs, and explanations together.

Advantages:

- Easy to learn.
- Excellent for EDA and Machine Learning.
- Supports charts and visualizations.

Example:

```
import pandas as pd
df = pd.read_csv("diabetes.csv")
df.head()
```

Google Colab

Description:

- Online version of Jupyter Notebook.
- Runs in a web browser without installation.

Advantages:

- Free to use.
- Accessible from anywhere.
- Supports GPU for Deep Learning.
- Easy sharing using Google Drive.

Visual Studio Code (VS Code)

Lightweight code editor developed by Microsoft.

Advantages:

- Fast and customizable.
- Supports Python extensions.
- Built-in terminal and Git integration.

Best For:

- Python programming
- Machine Learning projects
- Software development

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PyCharm

A full-featured Python IDE developed by JetBrains.

Advantages:

- Intelligent code completion.
- Powerful debugger.
- Easy project management.

Best For:

- Large Python applications
- Professional development

Spyder

Scientific IDE for Python.

Advantages:

- Variable Explorer.
- Integrated plots.
- Popular among researchers.

Anaconda Navigator

A graphical platform for Data Science.

Advantages:

- Easy package management.
- Launches Jupyter Notebook, Spyder, and other tools.
- Includes popular libraries such as NumPy, Pandas, and Matplotlib.

WHERE THIS PATH CAN TAKE YOU

Career Connection

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Data Analyst

Explores data and builds reports/dashboards

Data Scientist

Builds models to predict and explain outcomes

Data Engineer

Builds pipelines that move and store data

ML Engineer

Deploys machine learning models into production

AI Engineer

Builds intelligent systems using ML & deep learning

Business Analyst

Connects data insight with business strategy

Core skills across all roles: Python • Statistics • SQL • Communication • Domain Knowledge

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Data Science:

Extracting insights from data using statistics, programming, and machine learning.

The 4 V's:

Volume, Velocity, Variety, and Veracity explain why Big Data needs Data Science.

Six Fundamentals:

Collection, cleaning, analysis, visualization, ML, and decision-making.

Roles & Responsibilities:

From business understanding through to communicating results.

Toolbox:

Python/R, SQL, Pandas/NumPy, Matplotlib/Seaborn, Scikit-learn, Git.

Challenges:

Data quality, privacy, security, bias, and deployment all require care.

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EDA:

Initial analysis of data to understand its characteristics and identify issues.

Data Science Process:

Business problem through deployment and monitoring, as a continuous loop.

Methodology:

Seven structured steps from Business Understanding to Model Evaluation.

Why Python:

Simple, open-source, and backed by a complete Data Science library ecosystem.

IPython:

Enhanced interactive shell with auto-completion and magic commands.

Jupyter Notebook:

Interactive environment combining code, visualization, and documentation.

PRACTICE & ASSESSMENT

Practice Questions — Short Answer (2–5 Marks)

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ANSWER IN 2-5 SENTENCES

Q1. Define Data Science and list two of its applications.

Q2. What is Exploratory Data Analysis (EDA), and why is it important?

Q3. Differentiate between IPython and Jupyter Notebook.

Q4. List any four tools in a Data Scientist's toolbox.

Q5. Name the 4 V's of Big Data and briefly explain one of them.

Q6. What is the difference between a Data Scientist and a Data Analyst?

Practice Questions — Descriptive (5–10 Marks)

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ANSWER WITH DIAGRAMS/EXAMPLES WHERE RELEVANT

Q1. Explain the Data Science process from business problem to deployment with a diagram.

Q2. Describe the roles and responsibilities of a Data Scientist across the data lifecycle.

Q3. Discuss the Data Science methodology step-by-step, using a suitable example.

Q4. Explain why Python is preferred for Data Science, covering IPython and Jupyter Notebook.

Q5. Discuss any four challenges faced in Data Science, with a mitigation for each.

Q6. Explain EDA, its objectives, and any three EDA techniques with examples.

PRACTICE & ASSESSMENT

Practice Questions — Quick MCQ Recap

1. Which library is primarily used for numerical arrays in Python?

A. Pandas

B. NumPy

C. Seaborn

D. Flask

2. Which chart best shows the correlation between many variables at once?

A. Histogram

B. Box Plot

C. Heatmap

D. Pie Chart

3. Which of these is an unsupervised learning task?

A. Clustering

B. Regression

C. Classification

D. Time-series forecasting

4. In the methodology, which step comes right after Data Acquisition?

A. Modeling

B. Data Understanding

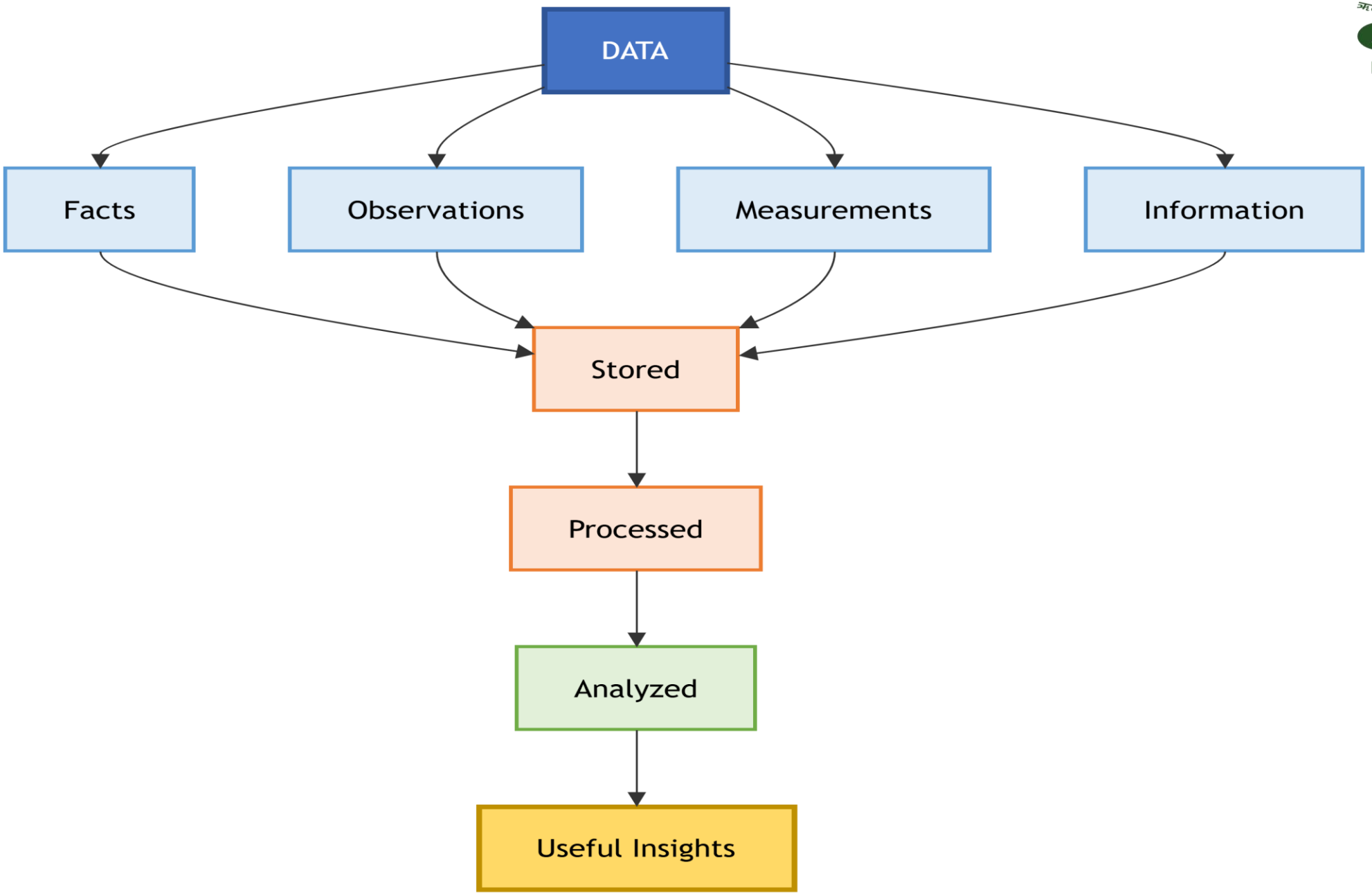
C. Model Evaluation

D. Business Understanding

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BASIC TYPES OF DATA IN Data Science

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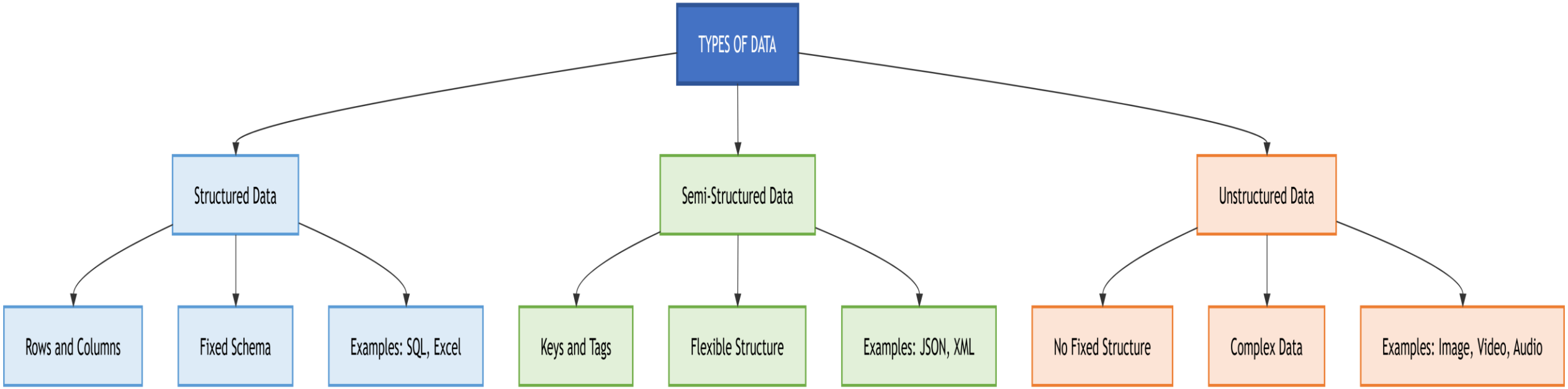
Data is a collection of facts, observations, measurements, or information that can be stored, processed, and analyzed to obtain useful knowledge.

Examples:

- Student name
- Student marks
- Attendance
- Photograph
- Video
- Audio
- Customer transaction
- Temperature reading

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Types of Data



Structured Data

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- Structured data is organized into rows and columns. Data represented in a tabular format is called structured data. It is easy to store, analyze, and visualize.

Student_ID	Age	Gender	Degree	Branch	CGPA	Internships	Projects	Coding_Sk	Communi	Aptitude_T	Soft_Skills	Certificati	Backlogs	Placement_Status
1048	22	Female	B.Tech	ECE	6.29	0	3	4	6	51	5	1	3	Not Placed
37820	20	Female	BCA	ECE	6.05	1	4	6	8	59	8	2	1	Not Placed
49668	22	Male	MCA	ME	7.22	1	4	6	6	58	6	2	2	Not Placed
19467	22	Male	MCA	ME	7.78	2	4	6	6	90	4	2	0	Placed
23094	20	Female	B.Tech	ME	7.63	1	4	6	5	79	6	2	0	Placed
8710	20	Male	BCA	ECE	7.99	1	4	5	7	84	6	2	0	Placed
24363	21	Male	MCA	ECE	7.5	1	4	6	4	71	8	2	1	Not Placed
27448	19	Male	B.Tech	ME	8	0	4	4	5	74	3	1	0	Not Placed
23663	24	Male	BCA	Civil	6.24	0	3	3	10	54	7	1	2	Not Placed
39336	23	Male	B.Sc	ME	7.08	2	4	7	6	59	6	2	2	Not Placed
21919	18	Female	MCA	Civil	5.42	0	2	1	7	42	5	0	1	Not Placed
11924	19	Male	MCA	ECE	7.48	2	5	9	8	67	6	3	0	Placed
11236	19	Male	B.Sc	CSE	7.58	1	5	8	1	80	8	3	2	Not Placed
31588	18	Male	B.Tech	ECE	6.74	2	4	7	6	73	8	2	0	Placed
11221	19	Female	MCA	ECE	6.11	1	3	5	5	57	5	1	2	Not Placed
6245	18	Male	BCA	ECE	6.79	0	3	5	5	81	6	2	0	Placed
41110	23	Female	B.Sc	IT	7.41	0	3	3	5	83	5	1	1	Not Placed
46279	18	Female	B.Sc	CSE	5.97	0	3	6	4	52	6	2	2	Not Placed
25157	23	Male	B.Tech	CSE	6.69	1	4	8	7	68	6	2	1	Placed
42829	20	Male	BCA	Civil	7.95	1	4	5	6	81	6	2	1	Placed
32822	18	Female	B.Tech	ME	7.3	1	4	7	8	70	7	2	0	Placed
10557	18	Female	MCA	ME	7.02	0	3	3	8	81	6	1	0	Not Placed
33595	23	Female	BCA	ECE	7.29	0	3	3	6	88	5	1	0	Not Placed
49018	21	Male	BCA	ME	7.69	1	3	4	5	54	9	1	0	Not Placed
2359	24	Male	BCA	CSE	8.47	2	5	9	3	80	4	3	0	Not Placed
16180	24	Female	B.Tech	ECE	7.97	1	4	7	5	89	6	2	1	Placed
19260	23	Male	B.Tech	ECE	7.58	2	4	5	4	66	6	2	2	Not Placed
26634	23	Female	MCA	Civil	7.08	2	4	7	7	83	4	2	0	Placed
9097	24	Female	B.Sc	CSE	6.89	1	4	8	5	69	5	2	0	Placed
19470	24	Male	B.Tech	ECE	7.49	2	4	5	3	80	4	2	0	Not Placed
32452	20	Male	BCA	IT	6.69	1	3	4	7	64	7	1	1	Not Placed
49309	21	Female	MCA	ME	6.29	1	3	3	4	55	5	1	2	Not Placed
32207	20	Female	B.Sc	Civil	8	0	4	5	7	74	6	2	0	Placed

Examples

- SQL databases
- Excel spreadsheets
- Banking transactions
- Student records
- Employee databases
- Inventory databases

```
import pandas as pd
data = {
    "Student_ID": [101, 102, 103],
    "Name": ["Amit", "Riya", "XY"],
    "Age": [20, 21, 20],
    "CGPA": [8.2, 8.8, 7.5]
}
df = pd.DataFrame(data)
print(df)
```

```
   Student_ID  Name  Age  CGPA
0         101  Amit   20   8.2
1         102  Riya   21   8.8
2         103    XY   20   7.5
```

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```
import pandas as pd
```

```
df=pd.read_csv("/content/diabetes1.csv")
```

```
df.head(2)
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Diabetes
0	6.0	148.0	72.0	35.0	125.0	33.6	0.627	50	1
1	1.0	85.0	66.0	29.0	125.0	26.6	0.351	31	0

```
df.tail(2)
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Diabetes
766	1.0	126.0	60.0	29.0	125.0	30.1	0.349	47	1
767	1.0	93.0	70.0	31.0	125.0	30.4	0.315	23	0

Semi-Structured Data

```
{
  "Name": "Rahul",
  "Age": 20,
  "Skills": ["Python", "SQL"]
}
```

- Semi-structured data is data that does not follow a fixed table structure but contains tags or key-value pair that provide some organization.
- Unlike structured data, it is not necessarily stored in a table.
- It uses keys, tags, attributes, or metadata to organize information.
- It has a flexible structure , different records can have different fields.
- Data are Self-describing , the data often tells us what each value represents.

Common formats : JSON ,XML ,HTML

- Common uses :
  - Web applications
  - Data exchange
  - Log files
  - Configuration files

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```python
import json
student = {
    "Student_ID": 101,
    "Name": "Rahul",
    "Age": 20,
    "Skills": ["Python", "SQL"]
}
json_data = json.dumps(student, indent=5)
print(json_data)
```

```
{
     "Student_ID": 101,
     "Name": "Rahul",
     "Age": 20,
     "Skills": [
          "Python",
          "SQL"
     ]
}
```

Unstructured Data

- Unstructured data is data that does not follow a predefined format or fixed schema such as rows and columns.

[8]
✓ 0s

```
text = """
1. Ram attended the Python class today.
2. He learned about Machine Learning.
3. He completed his assignment.Scored 95%
4. his CGPA is 99%
"""

print(text)
```

```
...
1. Ram attended the Python class today.
2. He learned about Machine Learning.
3. He completed his assignment.Scored 95%
4. his CGPA is 99%
```

- No fixed rows and columns :It cannot naturally be represented as a table.
- No predefined schema :We don't define fixed fields such as `Name`, `Age`, `CGPA`.
- Different forms of data : It can be:
    - Images
    - Videos
    - Audio
    - Natural-language text
    - Social media content
    - PDF/documents
- Difficult to analyze directly : Traditional database techniques cannot easily analyze it.
- Requires specialized techniques : We often use:
    - NLP → Text
    - Computer Vision → Images/Video
    - Speech Processing → Audio
    - Machine Learning / Deep Learning → Extract useful information

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Data Analytics

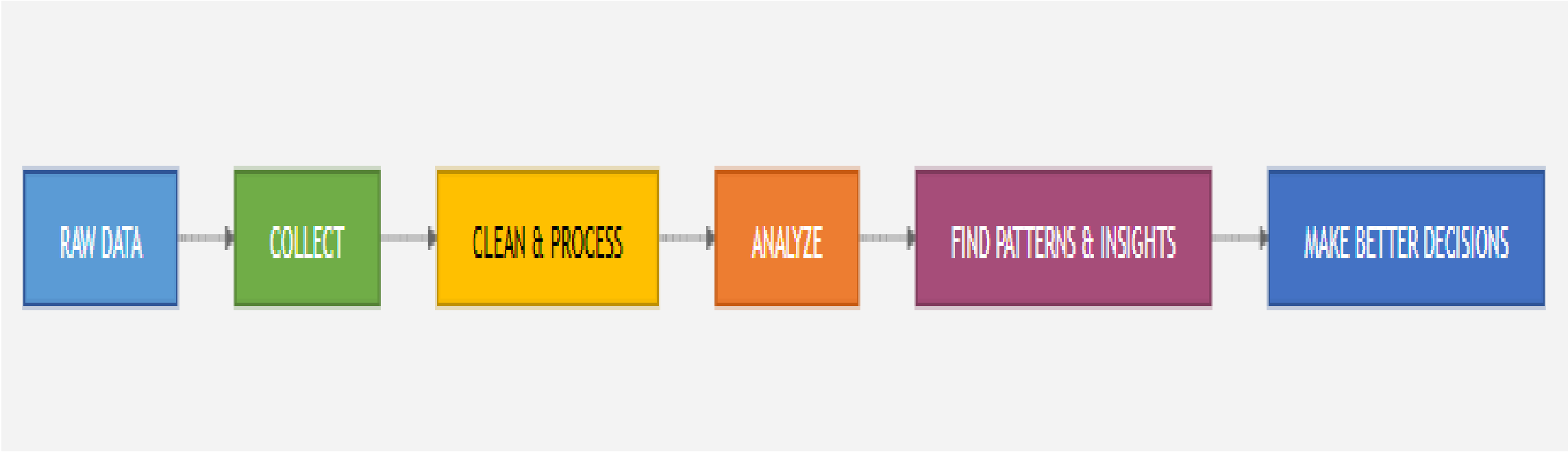
Data Analytics is the process of examining, processing, and interpreting data to discover meaningful patterns, insights, and information that help in making better decisions.

```
df.head(10)
```

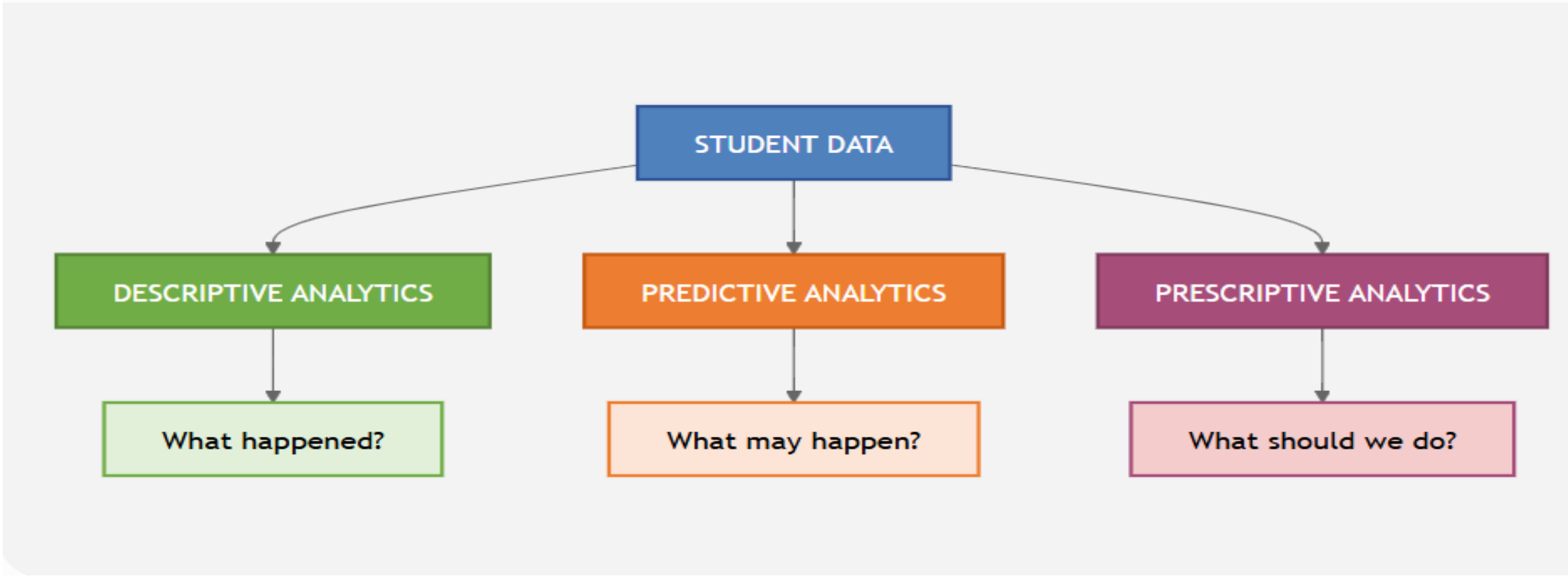
	Student_ID	Attendance (%)	Internal Test 1 (out of 40)	Internal Test 2 (out of 40)	Assignment Score (out of 10)	Daily Study Hours	Final Exam Marks (out of 100)
0	S1000	84	30	36	7	3	72
1	S1001	91	24	38	6	3	56
2	S1002	73	29	26	7	3	56
3	S1003	80	36	35	7	3	74
4	S1004	84	31	37	8	3	66
5	S1005	100	34	34	7	3	79
6	S1006	96	40	36	8	3	83
7	S1007	83	39	37	7	3	77
8	S1008	91	30	37	8	2	71
9	S1009	87	27	37	8	3	61

What happened? → Descriptive Analytics
What may happen? → Predictive Analytics
What should we do? → Prescriptive Analytics

Example:
Analyzing student attendance, assignment marks, and exam marks to understand student performance and decide which students need additional support



Data Analytics


STUDENT DATA
DESCRIPTIVE ANALYTICS
PREDICTIVE ANALYTICS
PRESCRIPTIVE ANALYTICS
What happened?
What may happen?
What should we do?

Descriptive Analysis(What has happened in the past?)

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- Come the word from Describe
- It is used to analyze historical data
- It aims to find pattern and relation ship in data
- Descriptive analytics uses statistical techniques to tell us what happened in the past by identifying patterns or trends.
- It helps us to understand "What has happened in the past?"

It can tell us:

- Average
- Maximum
- Minimum
- Total
- Count
- Percentage
- Frequency
- Trends
- Patterns

Common tools include:

- Tables
- Charts
- Graphs
- Reports
- Dashboards
- Statistical measures

- Descriptive Analytics is the process of collecting, summarizing, and analyzing historical data to understand what has happened in the past

Area	What Descriptive Analytics tells us
Revenue	How much revenue was generated last month/year?
Product Sales	Which product was sold the most? Which product had low sales?
Hospital Patients	How many patients visited? How many were admitted or discharged?
Clients	How many clients were acquired? How many targets were achieved?
Customer Feedback	How many customers were satisfied, neutral, or dissatisfied?
Customer Behavior	What products did customers purchase most frequently?
Employee Performance	Which employees achieved their targets? What was their sales/performance?

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("/content/Final_Marks_Data.csv")
print(df)
```

```
     Student_ID  Attendance (%)  Internal Test 1 (out of 40)  \
0         S1000              84                           30
1         S1001              91                           24
2         S1002              73                           29
3         S1003              80                           36
4         S1004              84                           31
...         ...             ...                          ...
1995      S2995              82                           31
1996      S2996              78                           38
1997      S2997              78                           30
1998      S2998              82                           29
1999      S2999              97                           32

      Internal Test 2 (out of 40)  Assignment Score (out of 10)  \
0                              36                             7
1                              38                             6
2                              26                             7
3                              35                             7
4                              37                             8
...                           ...                           ...
```

```
df.head(10)
```

	Student_ID	Attendance (%)	Internal Test 1 (out of 40)	Internal Test 2 (out of 40)	Assignment Score (out of 10)	Daily Study Hours	Final Exam Marks (out of 100)
0	S1000	84	30	36	7	3	72
1	S1001	91	24	38	6	3	56
2	S1002	73	29	26	7	3	56
3	S1003	80	36	35	7	3	74
4	S1004	84	31	37	8	3	66
5	S1005	100	34	34	7	3	79
6	S1006	96	40	36	8	3	83
7	S1007	83	39	37	7	3	77
8	S1008	91	30	37	8	2	71
9	S1009	87	27	37	8	3	61

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```python
#Find Average Marks
average = df["Final Exam Marks (out of 100)"].mean()
print("Average Marks:", average)
```

```
Average Marks: 64.855
```

```python
#Find Maximum Marks
maximum = df["Final Exam Marks (out of 100)"].max()
print("Maximum Marks:",maximum)
```

```
Maximum Marks: 100
```

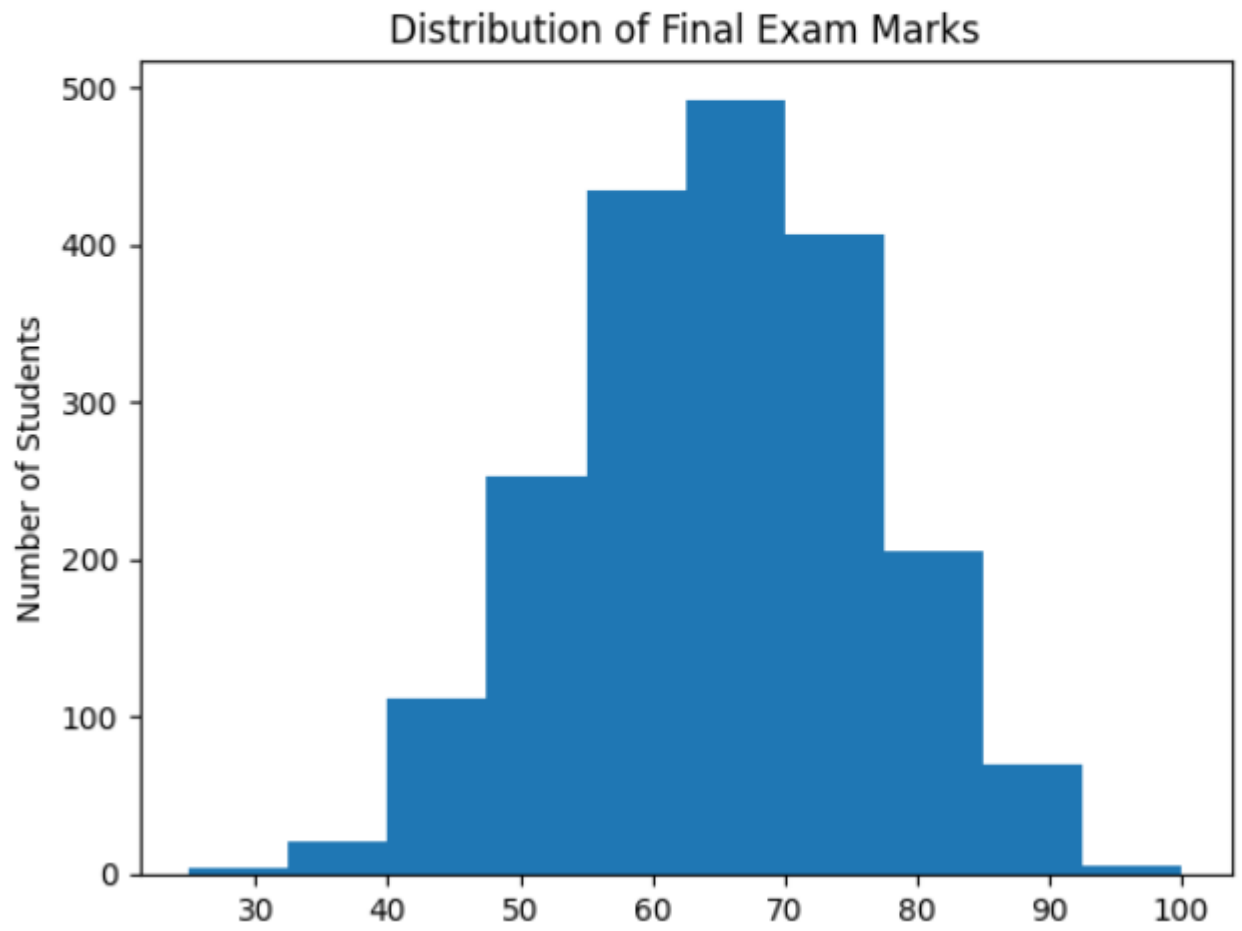
```python
#Find Minimum Marks
minimum = df["Final Exam Marks (out of 100)"].min()
print("Maximum Marks:",minimum)
```

```
Maximum Marks: 25
```

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```
plt.hist(df["Final Exam Marks (out of 100)"], bins=10)
plt.xlabel("Final Exam Marks")
plt.ylabel("Number of Students")
plt.title("Distribution of Final Exam Marks")

plt.show()
```



Predictive Analytics

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Predictive Analytics is the process of using historical data, statistical techniques, and machine learning algorithms to predict a future or unknown outcome.

[37] ✓ 0s

```
X = df[[
    "Attendance (%)",
    "Internal Test 1 (out of 40)",
    "Internal Test 2 (out of 40)",
    "Assignment Score (out of 10)"
]]
Y=df["Final Exam Marks (out of 100)"]
```

[38] ✓ 0s

```
X_train, X_test, y_train, y_test = train_test_split(X,Y,
    test_size=0.2,
    random_state=42
)
```

[40] ✓ 0s

```
model = LinearRegression()
model.fit(X_train, y_train)
```

▾ LinearRegression

```
LinearRegression()
```

[46] ✓ 0s

```
new_student = [[90, 40, 40, 8]]

prediction = model.predict(new_student)

print("Predicted Final Marks:", prediction[0])
```

```
Predicted Final Marks: 82.1801112063586
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
  warnings.warn(
```

Prescriptive Analytics

Prescriptive Analytics is the process of using data and predictions to recommend an appropriate action or decision.

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```
predicted_marks = prediction

if predicted_marks < 50:
    print("Recommendation: Provide additional classes")

elif predicted_marks > 50 and predicted_marks < 60:
    print("Recommendation: Give extra practice")

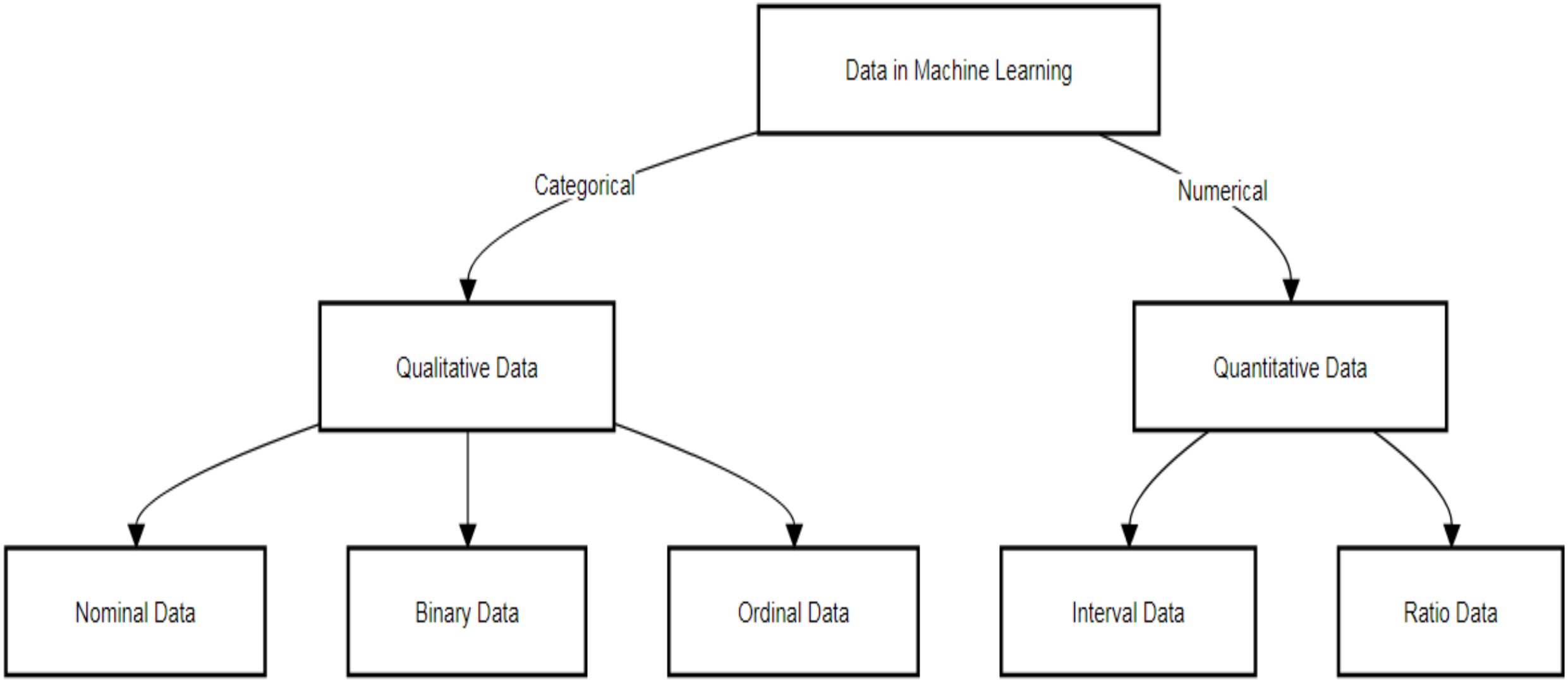
else:
    print("Recommendation: Continue regular learning")
```

```
Recommendation: Continue regular learning
```

- Prescriptive analytics takes predictive data to the next level. Now that you have an idea of what will likely happen in the future,
- what should you do about it? This type of analysis suggests various courses of action and outlines what the potential implications would be for each.

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Measurement & Scaling(Nominal, Ordinal ,Interval , Ratio)


Data in Machine Learning
Categorical
Numerical
Qualitative Data
Quantitative Data
Nominal Data
Binary Data
Ordinal Data
Interval Data
Ratio Data

Qualitative Data (Categorical Data)

- This type of data represents categories or labels that do not have a numerical meaning.
- It describes qualities, attributes, or characteristics of an entity.
- It is further divided into:
  - Nominal Data
  - Ordinal Data
  - Binary Data

Nominal Scale (Nominal Data)

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Nominal data is a type of qualitative (categorical) data that represents labels or names without any inherent order or ranking. The values in nominal data are used only for identification and classification purposes.

Characteristics of Nominal Data:

- **No natural ordering** (e.g., Red is not greater than Blue).
- **Cannot perform arithmetic operations** like addition or averaging.
- **Can only be compared for equality or inequality** (e.g., Male ≠ Female).
    - **Examples of Nominal Data:**
- **Colors:** Red, Blue, Green
- **Gender:** Male, Female, Other
- **Blood Group:** A, B, AB, O
- **Nationality:** American, Indian, Chinese
- **Modes of Transport:** Car, Bike, Bus, Train
- Statistical function such as mean, variance etc can also not be applied on nominal data
- However basic count is possible. For example, how many times B+ frequently occurring.

Blood Group Data Analysis

Consider a dataset of blood groups collected from students:

Blood Group Data: ["A+", "B+", "O-", "B+", "A+", "AB+", "B+", "O+", "A+", "B+"]

- **Count of Each Blood Group**
  - A+ → **3 times**
  - B+ → **4 times**
  - O- → **1 time**
  - O+ → **1 time**
  - AB+ → **1 time**
- **Most Common Blood Group (Mode)**
  - **B+ (Most Frequent)**

"Nominal scale is a level of measurement used to classify data into distinct categories or groups using names, labels, or codes, where the categories have no natural order or ranking."

Binary Data/Binary Scale

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- Binary data is a type of qualitative (categorical) data that consists of only two distinct categories. These categories usually represent opposite or mutually exclusive states, such as Yes/No, 0/1, True/False.
- For Example whether a patient smokes or does not smoke, using 1 for smokers and 0 for non-smokers.
- A binary attributes referred to as Boolean, if the two states correspond to true and false.
- A Binary attribute is Symmetric Binary Attribute:
- Both values of the binary attribute are equally important and carry the same weight.
- There is no preference or hierarchy between the two categories.
- Example: Gender (Male = 0, Female = 1)
  - Neither category is inherently more important than the other.
- Asymmetric Binary Attribute:
- One value is more significant than the other.
- Often occurs when one value is rare or meaningful while the other represents the absence of a condition.
- Example: Disease Test Result (Positive = 1, Negative = 0)
  - "Positive" (1) is more meaningful, while "Negative" (0) is just the absence of disease.

Ordinal Scale\Ordinal Data

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- Ordinal scale is a level of measurement in which data is classified into categories that have a meaningful order or ranking
- For example:

Customer satisfaction :Unhappy → Happy → Very Happy

Grades : A,B,C

A is higher than B, and B is higher than C.

Hardness: Soft → Hard → Very Hard

Quantitative Data

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- Quantitative data relates to information about the quantity or amount of an object; hence, it can be measured.
- For example, if we consider the attribute “marks”, it can be measured using a scale of measurement.
- Quantitative data is also termed as numeric data.
- There are two types of quantitative data:
  - Interval data
  - Ratio data

Interval-Scaled Data

- Interval data is numerical data where the values have a proper order and the difference between two values is meaningful and equal.
- For example, consider temperature:10°C → 20°C → 30°C → 40°C
- 40°C is greater than 30°C. 30°C is greater than 20°C,and The values have an order.
- An ideal example of interval data is Celsius temperature.
- In interval data, the difference between values remains equal.
- For example, the difference between 12°C and 18°C is 6°C.
- Similarly, the difference between 15.5°C and 21.5°C is also 6°C.
- So the difference between values is meaningful and remains equal.
- Other examples: Date, time, IQ score, etc.

Ratio Scale\Ratio Data

- Ratio scale or Ration data is a data in which values have a meaningful order, equal differences between values, and a true zero point.
- For Example: Weight
  - Order: 20 kg < 40 kg < 60 kg
  - Difference: 40 − 20 = 20 kg
  - True zero: 0 kg means no weight

Other examples:

- Height
- Weight
- Age
- Distance
- Income
- Time duration
- Number of students

House price prediction

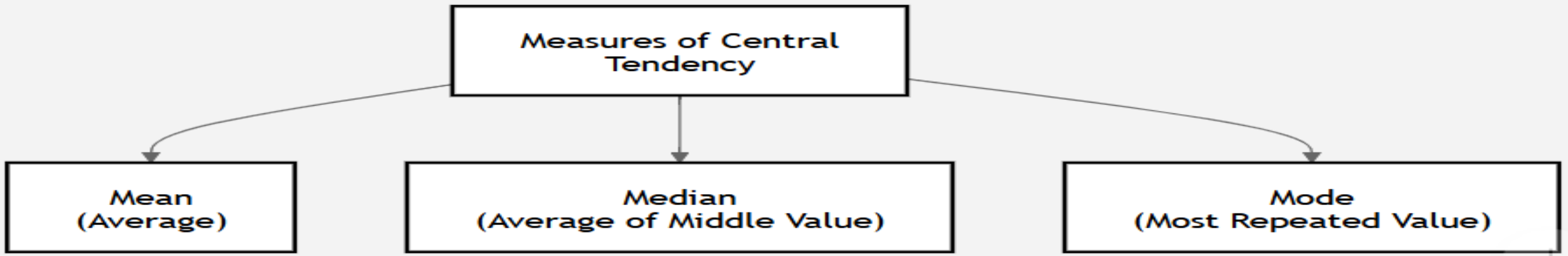
Area → 1000, 1500, 2000 sq.ft

Number of rooms → 2, 3, 4

Price → ₹30 lakh, ₹50 lakh, ₹70 lakh

Measures of Central Tendency

- Central Tendency in Statistics is a concept used to identify the central or average value within a data set.
- It provides a single value that represents the typical or central value of the data distribution.
- The main three measures of central tendency are the:
  - Mean(Average)
  - median(Middle value)
  - mode(Most repeated value)



Mean (Arithmetic Mean)

- The mean is the most common measure of central tendency.
- In simple words, mean means average.
- It gives us one value that represents the data.
- The mean is the average value of a dataset, obtained by adding all observations and dividing by the number of observations.

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

Using sigma notation:$= \frac{\Sigma x}{n}$

$$\bar{x} = \frac{\sum x}{n}$$

Data: 10, 20, 30, 40

$$\text{Mean} = \frac{10 + 20 + 30 + 40}{4} = 25$$

With an Outlier

Data: 10, 20, 30, 200

$$\text{Mean} = \frac{10 + 20 + 30 + 200}{4} = 65$$

Advantages

- Easy to calculate and understand
- Uses all values in the dataset
- Suitable for normally distributed data

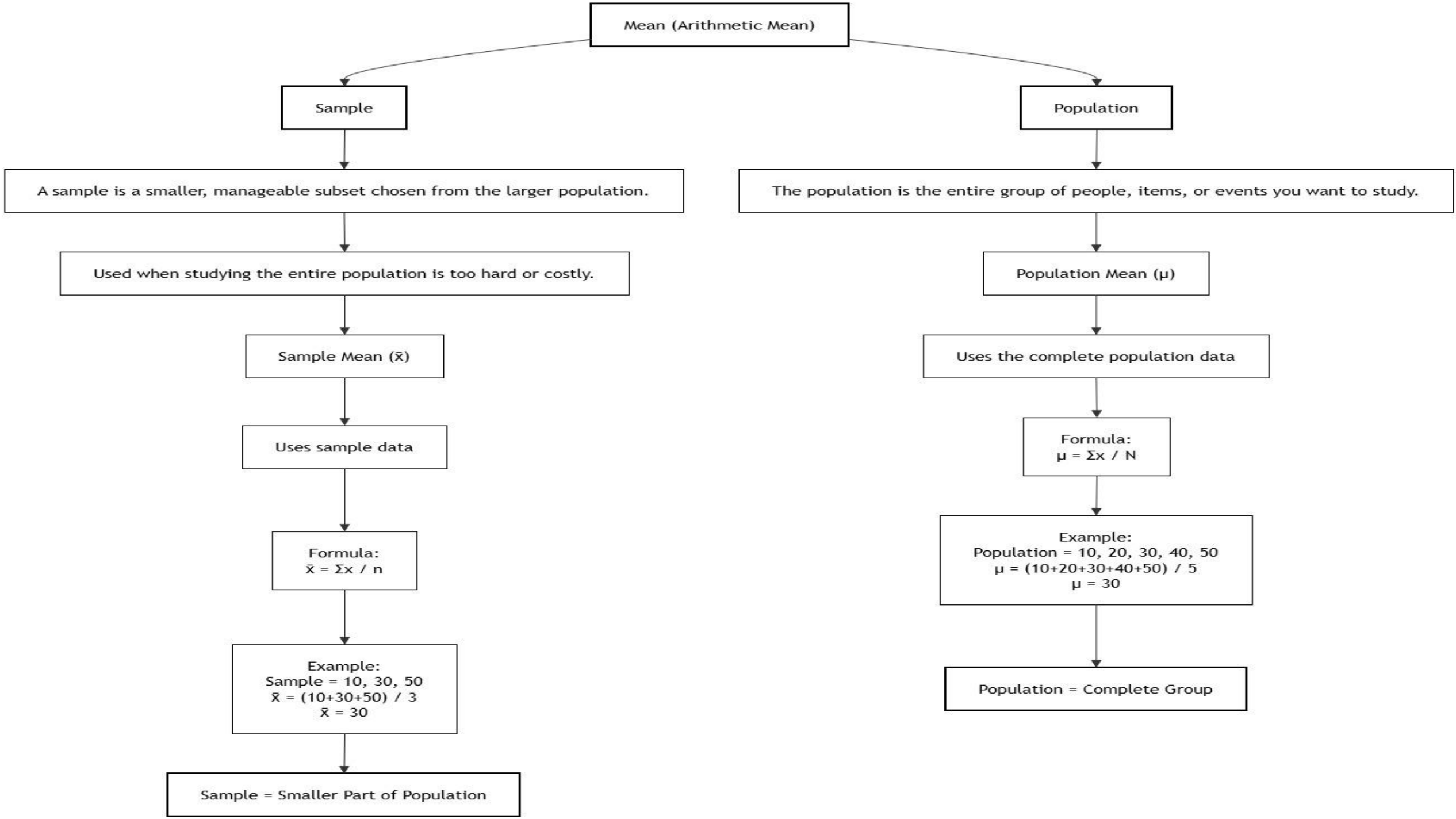
Disadvantages

- Highly sensitive to outliers
- Not suitable for skewed data
- May give unrealistic results

central tendency can be applied on two types of data

- Population data(The population is the entire group of people, items, or events you want to study.
- Example: A college has 5,000 students. All 1,000 students are the population.

- Sample Data: sample is a smaller, manageable subset chosen from the larger population. Researchers use it when studying the entire population is too hard or costly.


Mean (Arithmetic Mean)
Sample
Population
A sample is a smaller, manageable subset chosen from the larger population.
The population is the entire group of people, items, or events you want to study.
Used when studying the entire population is too hard or costly.
Population Mean (μ)
Sample Mean (x̄)
Uses the complete population data
Uses sample data
Formula:
μ = Σx / N
Formula:
x̄ = Σx / n
Example:
Population = 10, 20, 30, 40, 50
μ = (10+20+30+40+50) / 5
μ = 30
Example:
Sample = 10, 30, 50
x̄ = (10+30+50) / 3
x̄ = 30
Population = Complete Group
Sample = Smaller Part of Population

Median

- The median is the middle value of a dataset when arranged in ascending or descending order.

Case 1: if the dataset have Odd number of values:10, 15, 18, 17, 7

Step 1: Arrange the data :7,10,15,17,18

Step 2: Find the middle value: There are 5 values, so 15 is in the center.

$$Median = \text{value at position } \frac{n+1}{2}$$

Case 2: if the dataset have Even number of values:10, 40, 20, 30

Step 1: Arrange the data:10,20,30,40

Step 2: Take their average: Median = (20 + 30) / 2 = 25

$$\text{Median} = \frac{\text{Middle two values}}{2}$$

Advantages

- Not affected by outliers
- Good for skewed data

Disadvantages

- Does not use all values
- Harder for large data

Mode

- Mode is the value that occurs most frequently.

$$\text{Mode} = \text{Most frequent value}$$

For example:

Data: 2, 4, 4, 6
Mode = 4

Advantages

- Best for categorical data
- Easy to identify

Disadvantages

- May be no mode or many modes
- Not always representative

(1) Find the mean, median and mode for the following data:

25, 37, 31, 32, 32, 29, 30, 32, 29, 34

(2) Find the mean, median, mode, and range for the given data:

190, 153, 168, 179, 194, 153, 165, 187, 190, 170, 165, 189, 185, 153, 147, 161, 127, 180

(3) Find the Median of the data 25, 12, 5, 24, 15, 22, 23, 25

(4)A teacher recorded the following test scores for her class: [85, 78, 92, 88, 84, 90, 79, 81, 87, 93, 78, 84].

1. Calculate the mean score.
2. Determine the median score.
3. Identify the mode(s) if any.

1. Find the mean, median and mode for the following data: 25, 37, 31, 32, 32, 29, 30, 32, 29, 34

(Ans: Mean: 31.1 units, Median: 31.5 units, Mode: 32 units)

2. Find the mean, median and mode for the following data:

No. of vehicles	0	1	2	3	4	5
No. of families	3	13	18	10	5	1

(Ans: Mean: 2.04 vehicles per family, Median: 2 vehicles per family, Mode: 2 vehicles per family)

3. Given below is the data about the fruit boxes received by a fruit vendor. Find arithmetic mean.

Weight in Kg.	10	11	12	14	15
No. of boxes	4	6	6	5	2

(Ans: 12.08 Kg.)

Vehicles x	Families f	fx
0	3	$0 \times 3 = 0$
1	13	$1 \times 13 = 13$
2	18	$2 \times 18 = 36$
3	10	$3 \times 10 = 30$
4	5	$4 \times 5 = 20$
5	1	$5 \times 1 = 5$
Total	50	104

In case of frequency distribution

$$\text{Mean} = \frac{\sum fx}{\sum f}$$

Vehicles x	Frequency f	Cumulative Frequency
0	3	3
1	13	16
2	18	34
3	10	44
4	5	49
5	1	50

50

If N=50 ➔50/2=25
Cumulative frequency just greater than N/2 is 34,and value of x is 2 corresponding to 34,therefore median is 2.

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Example 2·1. *(a) Find the arithmetic mean of the following frequency distribution:*

x :	1	2	3	4	5	6	7
f :	5	9	12	17	14	10	6

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Solution. *(a)*

x	f	fx
1	5	5
2	9	18
3	12	36
4	17	68
5	14	70
6	10	60
7	6	42
	73	299

$$\therefore \quad \bar{x} = \frac{1}{N} \Sigma f x = \frac{299}{73} = 4{\cdot}09$$

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(b) Calculate the arithmetic mean of the marks from the following table :

Marks	:	*0–10*	*10–20*	*20–30*	*30–40*	*40–50*	*50–60*
No. of students	:	*12*	*18*	*27*	*20*	*17*	*6*

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(b)

Marks	*No. of students* (*f*)	*Mid - point* (*x*)	*fx*
0–10	12	5	60
10–20	18	15	270
20–30	27	25	675
30–40	20	35	700
40–50	17	45	765
50–60	6	55	330
Total	100		2,800

Arithmetic mean or $\bar{x} = \frac{1}{N} \Sigma f x = \frac{1}{100} \times 2{,}800 = 28$

Calculate the mean of the following frequency distribution

step-Deviation Method

Class interval	Frequency f	Mid value(x)	d=x-A/h	fd=f*d
0–8	8	4	(4-28)/8=-3	-24
8–16	7	12	(12-28)/8=-2	-14
16–24	16	20	(20-28)/8=-1	-16
24–32	24	28	(28-28)/8=0	0
32–40	15	36	(36-28)/8=1	15
40–48	7	44	(44-28)/8=2	14
Total	77			-25

- If I use simple method to calculate fx then calculation will be large because
- Choose a convenient mid-value:

$$4,\ 12,\ 20,28,\ 36,\ 44$$

We choose :A=28(assumed mean , our starting point)

$$\bar{x} = A + h\frac{\sum fd}{N}$$

=28+8(-25/77)

=28-200/77=25.404

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(b) Calculate the arithmetic mean of the marks from the following table :

Marks	:	*0–10*	*10–20*	*20–30*	*30–40*	*40–50*	*50–60*
No. of students	:	*12*	*18*	*27*	*20*	*17*	*6*

Marks	Frequency f	Mid-value x	$d = \frac{x - 35}{10}$	fd
0–10	12	5	−3	−36
10–20	18	15	−2	−36
20–30	27	25	−1	−27
30–40	20	35	0	0
40–50	17	45	1	17
50–60	6	55	2	12
Total	100			−70

$$\bar{x} = A + h\frac{\sum fd}{N}$$

=35+10(-70 /100)
=35-700/100
=35-7
=28

(iii) The corresponding value of x is median.

Example 2·5. *Obtain the median for the following frequency distribution:*

x :	1	2	3	4	5	6	7	8	9
f :	8	10	11	16	20	25	15	9	6

x	f	cf
1	8	8
2	10	18
3	11	29
4	16	45
5	20	65
6	25	90
7	15	105
8	9	114
9	6	120
Total	120	

Example 2·6. *Find the median wage of the following distribution :*

Wages (in Rs.) :	20—30	30—40	40—50	50—60	60—70
No. of labourers :	3	5	20	10	5

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Wages (₹)	No. of labourers f	Cumulative frequency (cf)
20–30	3	3
30–40	5	8
40–50	20	28
50–60	10	38
60–70	5	43
Total	43	

N=43

$$\frac{N}{2} = \frac{43}{2} = 21.5$$

The first cf greater than 21.5 is 28, so the median class = 40–50.

Median Formula for Grouped Data

$$\text{Median} = L + \frac{\frac{N}{2} - cf}{f} \times h$$

Where:

- L = Lower limit of median class
- N = Total frequency
- cf = Cumulative frequency **before** median class
- f = Frequency of median class
- h = Class width

$$Median = L + \frac{N/2 - cf}{f} \times h$$

$$= 40 + \frac{21.5 - 8}{20} \times 10$$

$$= 40 + \frac{13.5}{20} \times 10$$

$$= 40 + 6.75$$

$$\boxed{Median = 46.75}$$

Find the median height:165.75

Height (cm)	No. of students
140–150	4
150–160	9
160–170	20
170–180	12
180–190	5

Find the median income:34,000

Income (₹)	No. of families
10,000–20,000	7
20,000–30,000	13
30,000–40,000	25
40,000–50,000	10
50,000–60,000	5

Find the mode for the following distribution :

Class interval	0–10	10–20	20–30	30–40	40–50	50–60	60–70	70–80
Frequency f	5	8	7	12	28	20	10	10

Maximum frequency is 28 so,
Model class=40-50
l=40 , f_1=28, f_0=12, f_2=20, h=10

Formula for Mode of Grouped Data

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

Where:

- l = lower limit of modal class
- f_1 = frequency of modal class
- f_0 = frequency of class **before** modal class
- f_2 = frequency of class **after** modal class
- h = class width

$$Mode = 40 + \frac{28 - 12}{2(28) - 12 - 20} * 10$$

$$= 40 + \frac{16}{56-32} * 10$$

$$= 40 + \frac{16}{24} * 10$$

$$= 40 + 6.67 = 46.67$$

Class Interval	Frequency
0-10	5
10-20	8
20-30	12
30-40	20
40-50	14
50-60	6

35.71

Class Interval	Frequency
0-10	7
10-20	12
20-30	18
30-40	30
40-50	22
50-60	10

46

Highest frequency class is Modal class

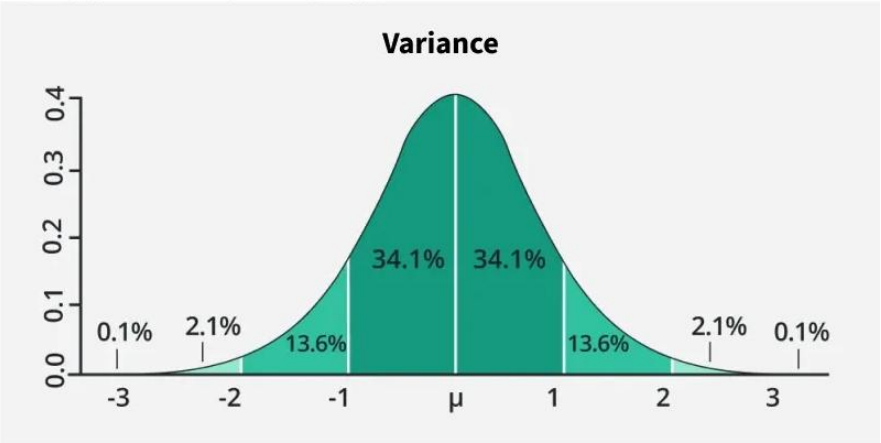
Class Interval	Frequency
0-10	4
10-20	11
20-30	19
30-40	32
40-50	24
50-60	9

33.81

Variance

- Variance tells us how much the data values are spread around the mean.
- Variance is a number that tells us how spread out the values in a data set are from the mean .
- It shows whether the numbers are close to the average or far away from it.
- If the variance is small, it means most numbers are close to the mean.
- If the variance is large, it means the numbers are spread out more widely.
- A higher variance indicates greater variability, meaning the data is spread, while a lower variance suggests the data points are closer to the mean.

The following image depicts the variance in a normal distribution, illustrating how data points are spread around the mean (μ).



Variance

Population Variance:

$$\sigma^2 = \frac{\sum (X - \mu)^2}{N}$$

Where:

- X= each data value
- μ= population mean
- N= total number of values
- σ^2= population variance

Sample Variance:

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

Where:

- x= each sample value
- $\bar{x}$= sample mean
- n= sample size
- s^2= sample variance

Steps to Calculate Variance

In general, variance means population standard variance. The steps to calculate the variance of a given set of values are,

- **Step 1:** Calculate the mean of the observation using the formula (Mean = Sum of Observations/Number of Observations)
- **Step 2:** Calculate the squared differences of the data values from the mean. (Data Value - Mean)^2
- **Step 3:** Calculate the average of the squared differences of the given values, which is called the variance of the data set.

(Variance = Sum of Squared Differences / Number of Observations)

Find the population variance of the data:5, 7, 9, 10, 14, 15

$$\text{Variance} = \frac{\sum (X - \mu)^2}{N}$$

Step 1: Find the Mean

$$\mu = \frac{5 + 7 + 9 + 10 + 14 + 15}{6}$$

$$\mu = \frac{60}{6} = \boxed{10}$$

So, Mean = 10.

Step 2: Find squared differences

X	$X - \mu$	$(X - \mu)^2$
5	5 − 10 = −5	25
7	7 − 10 = −3	9
9	9 − 10 = −1	1
10	10 − 10 = 0	0
14	14 − 10 = 4	16
15	15 − 10 = 5	25
Total		76

Step 3: Calculate Variance

$$\text{Variance} = \frac{76}{6}$$

$$\boxed{\text{Population Variance} = 12.67}$$

Calculate the sample variance of:5, 7, 9, 10, 14, 15

Ans:15.2

Calculate the sample variance for the following data:

X	10	20	30	40
Frequency (f)	2	3	4	1

$$s^2 = \frac{\sum f(X - \bar{x})^2}{n - 1}$$

Find Mean

x	f	fx
10	2	20
20	3	60
30	4	120
40	1	40
Total	10	240

Find squared difference

x	f	$x-\bar{x}$	$(x-\bar{x})^2$	$f(x-\bar{x})^2$
10	2	−14	196	392
20	3	−4	16	48
30	4	6	36	144
40	1	16	256	256
Total	10			840

$$\text{Mean}(\bar{x}) = \frac{240}{10} = 24$$

$$n = \Sigma f = 2+3+4+1 = 10$$

Sample Variance =

$$s^2 = \frac{\Sigma f (x-\bar{x})^2}{n-1}$$

$$s^2 = \frac{840}{9} = 93.33$$

Standard deviation

- Standard deviation is a measure of dispersion. It tells us how much the data values are spread around the mean.
- Standard deviation tells us how far the values are from the mean.
- We use standard deviation to understand whether the data values are:
    - Close to the mean → Small SD
    - Far from the mean → Large SD

Example

Data 1: 48, 49, 50, 51, 52

Values are close together then Small SD

Data 2: 10, 30, 50, 70, 90

Values are far apart then Large SD

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Population Standard Deviation

$$SD = \sqrt{\frac{\sum (X - \mu)^2}{N}}$$

Where:

- X= each value
- μ= population mean
- N= total number of values

Sample Standard Deviation

$$SD = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Where:

- x= each sample value
- $\bar{x}$= sample mean
- n= number of values

Relationship Between Variance and SD

$$\boxed{SD = \sqrt{Variance}}$$

Variance=SD2

$$Variance = 12.67$$

$$SD = \sqrt{12.67}$$

=3.56

Measure of Dispersion

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- Measure of dispersion tells us how much the data values are spread or scattered from each other and from the central value (mean, median, etc.).
- It tells us whether the data values are close together or far apart.

For example:

- 10, 11, 12, 13, 14 → values are close → less dispersion
- 2, 10, 20, 30, 50 → values are far apart → more dispersion

Two important measures are:

1. Range
2. IQR (Interquartile Range)

Range

Range is the difference between two extreme observation of the distribution .

$$\boxed{\text{Range} = \text{Largest value} - \text{Smallest value}}$$

If A is the largest observation and B is the smallest observation: $\boxed{\text{Range} = A - B}$

Example:

Data: 10, 15, 20, 25, 30

Largest value = 30

Smallest value = 10

$\text{Range} = 30 - 10 = 20$

Range tells us how spread out the data is from the smallest value to the largest value.

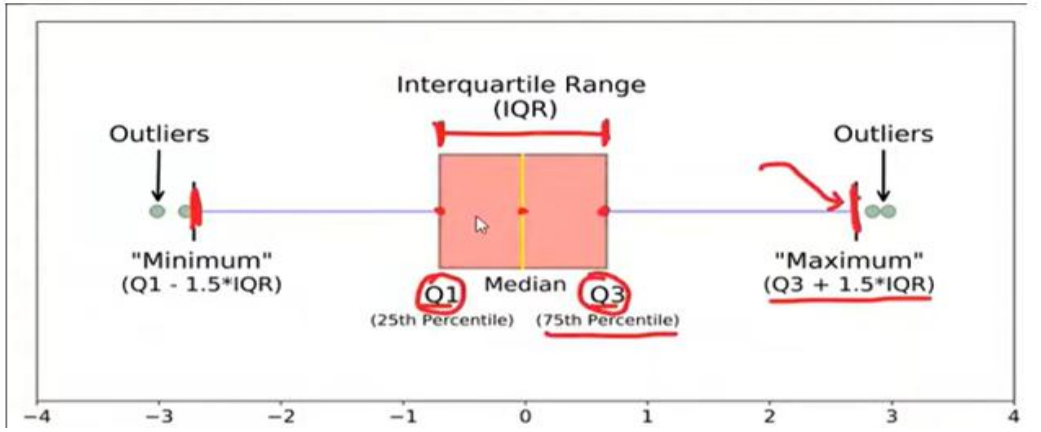
IQR (Interquartile Range)

- Interquartile Range (IQR) is a measure of dispersion that shows the spread of the middle 50% of the data.
- It is the difference between the third quartile (Q_3) and the first quartile (Q_1).

Formula:

$$IQR = Q_3 - Q_1$$

- Q_1 = Lower Quartile
- Q_3 = Upper Quartile



Q1 (First Quartile – 25%)
25% of data lies below Q1
Left edge of the box

Median (Q2 – 50%)
Middle value of the dataset
Shown as a line inside the box

Q3 (Third Quartile – 75%)
75% of data lies below Q3

- IQR Represents the middle 50% of the data
- Length of the box = IQR

IQR tells us how much the middle 50% of the data is spread out.

Find the Interquartile Range (IQR) of the following data:

25, 12, 35, 18, 40, 22, 15, 30, 28, 20

Solution:

Step 1: Arrange the Data in Ascending Order: 12, 15, 18, 20, 22, 25, 28, 30, 35, 40

Step2: Find Q_2(Median) value is in even number then Q2= (22+25)/2=23.5

Step 3: Divide the data into two halves

Lower half:12, 15, 18, 20, 22

Upper half:25, 28, 30, 35, 40

Step 4: Find Q_1,$Q_1$is the middle value of the lower half.12, 15, 18, 20, 22

$$Q_1 = 18$$

Step 5:Find Q_3,$Q_3$is the middle value of the upper half.25, 28, 30, 35, 40

$$Q_3 = 30$$

Step 6: Calculate IQR

Formula:

$$IQR = Q_3 - Q_1$$

$$IQR = 30 - 18 \ = 12$$

(1) The response time (in milliseconds) of a computer system is recorded as:

42, 35, 50, 28, 45, 60, 38, 55, 32, 48, 52, 40

Calculate:

1. Q_1
2. Q_2(Median)
3. Q_3
4. Interquartile Range (IQR)

(2) The following data represents the power consumption (in watts) of 11 IoT devices:

120, 135, 110, 150, 125, 140, 160, 130, 115, 145, 155

Find the:

1. First quartile Q_1
2. Third quartile Q_3
3. Interquartile Range

The following data represents the **power consumption (in watts)** of 11 IoT devices:

120, 135, 110, 150, 125, 140, 160, 130, 115, 145, 155

Find the:

1. First quartile Q_1
2. Third quartile Q_3
3. Interquartile Range

Outlier Detection

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•Largest and smallest value within the allowed range
$Q3 + 1.5 \times IQR$/Q1-1.5*IQR

5	The monthly salaries (in multiple of thousand rupees) of employees are: 25, 28, 32, 35, 37, 38, 40, 42, 45, 60 (i) Calculate the **IQR**. (ii) Using **IQR**, identify any outliers.	3	5	A	P

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Identify Outliers Using IQR

We have to calculate:

Lower Boundary

$$Q_1 - 1.5(IQR)$$
$$= 32 - 1.5(10)$$
$$= 32 - 15 = 17$$

Upper Boundary

$$Q_3 + 1.5(IQR)$$
$$= 42 + 1.5(10)$$
$$= 42 + 15 = 57$$

Therefore, the acceptable range is:

$$\boxed{17 \leq X \leq 57}$$

Any value below 17 or above 57 is an outlier.The data contains 60, which is greater than 57.60 is an outlier

The monthly salaries (₹ thousand) of 9 employees are:

25, 28, 32, 35, 37, 40, 42, 45, 60

Find:

1. Q_1
2. Q_3
3. IQR
4. Outliers using the IQR method.

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Arrange the data:25, 28, 32, 35, 37, 40, 42, 45, 60

Find the Median (Q_2)The middle value is:25, 28, 32, 35, [37], 40, 42, 45, 60

So: $Q_2 = 37$

Do not include the median when finding $Q_1$and Q_3.

Find Q_1Take the values below the median:25, 28, 32, 35

There are 4 values, so take the average of the middle two: $Q_1 = \frac{28+32}{2} = 30$

Find Q_3Take the values above the median:40, 42, 45, 60

$$Q_3 = \frac{42+45}{2} = 43.5$$

Find IQR,$IQR = Q_3 - Q_1$,$IQR = 43.5 - 30$=13.5

Find Outliers : Lower Boundary:$Q_1 - 1.5(IQR) = 30 - 1.5(13.5) = 30 - 20.25 = 9.75$

Upper Boundary:$Q_3 + 1.5(IQR) = 43.5 + 1.5(13.5) = 43.5 + 20.25 = 63.75$

So acceptable values are between:

$$\boxed{9.75 \text{ and } 63.75}$$

Our largest value is 60, which is below 63.75.No outlier

Handling Missing Data

- Missing data occurs when some values are not present in one or more rows or columns of a dataset.
- These missing values may contain important information that is required for data analysis and accurate prediction.
- In Python/Pandas, missing values are commonly represented by NaN (Not a Number).
- If missing values are not handled properly, they can affect the analysis, model training, and prediction results.
- Therefore, we need to identify and handle missing values before performing analysis or building a machine learning model.
- There are many methods for handling missing data.
- The appropriate method should be selected according to the type of data, amount of missing data, and requirement of the analysis.

Common methods of missing-value handling

- Delete rows/columns
- Mean imputation
- Median imputation
- Mode imputation
- Filling with Constant Value
- Forward Fill (ffill)
- Backward Fill (bfill)

```
Student = {
    'S_Name': ['Amit', 'Reena', 'Shyam', 'Ravi','abc','xyz','wqr','AAA','BBB','CCC'],
    'S_Age': [16,17,18,np.nan,20,21,23,20,np.nan,19],
    'Marks': [90, 70, 55, 56,30,np.nan,54,99,np.nan,78]
}
```

```
df=pd.DataFrame(Student)
```

```
df
```

	S_Name	S_Age	Marks
0	Amit	16.0	90.0
1	Reena	17.0	70.0
2	Shyam	18.0	55.0
3	Ravi	NaN	56.0
4	abc	20.0	30.0
5	xyz	21.0	NaN
6	wqr	23.0	54.0
7	AAA	20.0	99.0
8	BBB	NaN	NaN
9	CCC	19.0	78.0

- Ravi Age is missing.
- xyz Marks are missing.

Why Does Missing Data Occur?

Missing data can occur for many reasons:

1. **Non-response :** A participant does not answer a question.**Example :** A person does not provide their income.

2. **Equipment Failure :** A device or sensor fails to record information.

3. **Participant Withdrawal :** A participant leaves a study before completing it.

4. **Data Entry Error :** A value is accidentally left blank during data entry.

5. **Data Collection Problems :** A question or measurement is skipped.

Nature of Missingness

There are three important types:

1. MCAR(Missing Completely At Random)
2. MAR(Missing At Random)
3. MNAR(Missing Not At Random)

MCAR

- MCAR means that the missing values occur purely by chance , without any specific reason or pattern.
- Data is considered Missing Completely At Random (MCAR) if the probability of a data point being missing is unrelated to either the observed data or the missing data itself. In other words, there is no discernible pattern or reason behind the missingness.

Example : A survey where a random glitch in the system causes some responses to be lost, regardless of the respondent's answers or characteristics. During data entry, a technician accidentally skips random entries without any bias.

Why & How it Occurs?

- MCAR often occurs due to random, external factors such as:
  - Technical errors in data collection (e.g., random network failures).
  - Random accidents during manual data entry.

How to Solve?

- If data is MCAR, the missingness is truly random, and analysis is less biased. Common approaches include:
  - **Deletion Methods**: Remove rows or columns with missing data if the proportion of missingness is low.
  - Example: If only 1-2% of data points are missing, deletion might be acceptable.
  - **Imputation Techniques** : Use simple imputation methods like mean, median, or mode.

```
...
```

	S_Name	S_Age	Marks
0	Amit	16.0	90.0
1	Reena	17.0	70.0
2	Shyam	18.0	55.0
3	Ravi	NaN	56.0
4	abc	20.0	30.0
5	xyz	21.0	NaN
6	wqr	23.0	54.0
7	AAA	20.0	99.0
8	BBB	NaN	NaN
9	CCC	19.0	78.0

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MAR(Missing At Random)

- The probability of missingness is related to other observed variables.
- Data is Missing At Random (MAR) when the probability of missingness depends on the observed data but not on the missing data itself.
- In simpler terms, the missingness is systematic and can be explained by other variables in the dataset.

Example : Suppose income is more frequently missing for younger participants.Here, missingness in Income is related to Age, which is observed.

Age	Income
22	NaN
25	NaN
45	80,000
50	90,000

Why & How it Occurs?

- MAR typically arises due to relationships between observed variables and missingness:
  - Certain subgroups or patterns influence the likelihood of missing data.
  - Respondent behaviors or preferences are influenced by factors measured in the dataset.

How to Solve?

- Handling MAR requires understanding the relationships between variables:
  - **Imputation Techniques**:
    - Use conditional imputation based on observed variables. For example, impute income based on age, education, and occupation.
    - K-Nearest Neighbors (KNN): Impute missing values based on the most similar observations in the dataset.
    - Regression Imputation: Use regression models to predict missing values based on correlated observed variables.

Missing Not At Random (MNAR)

- Data is Missing Not At Random (MNAR) when the missingness depends on the value of the missing data itself.
- This introduces bias because the missing values are systematically different from the observed values.

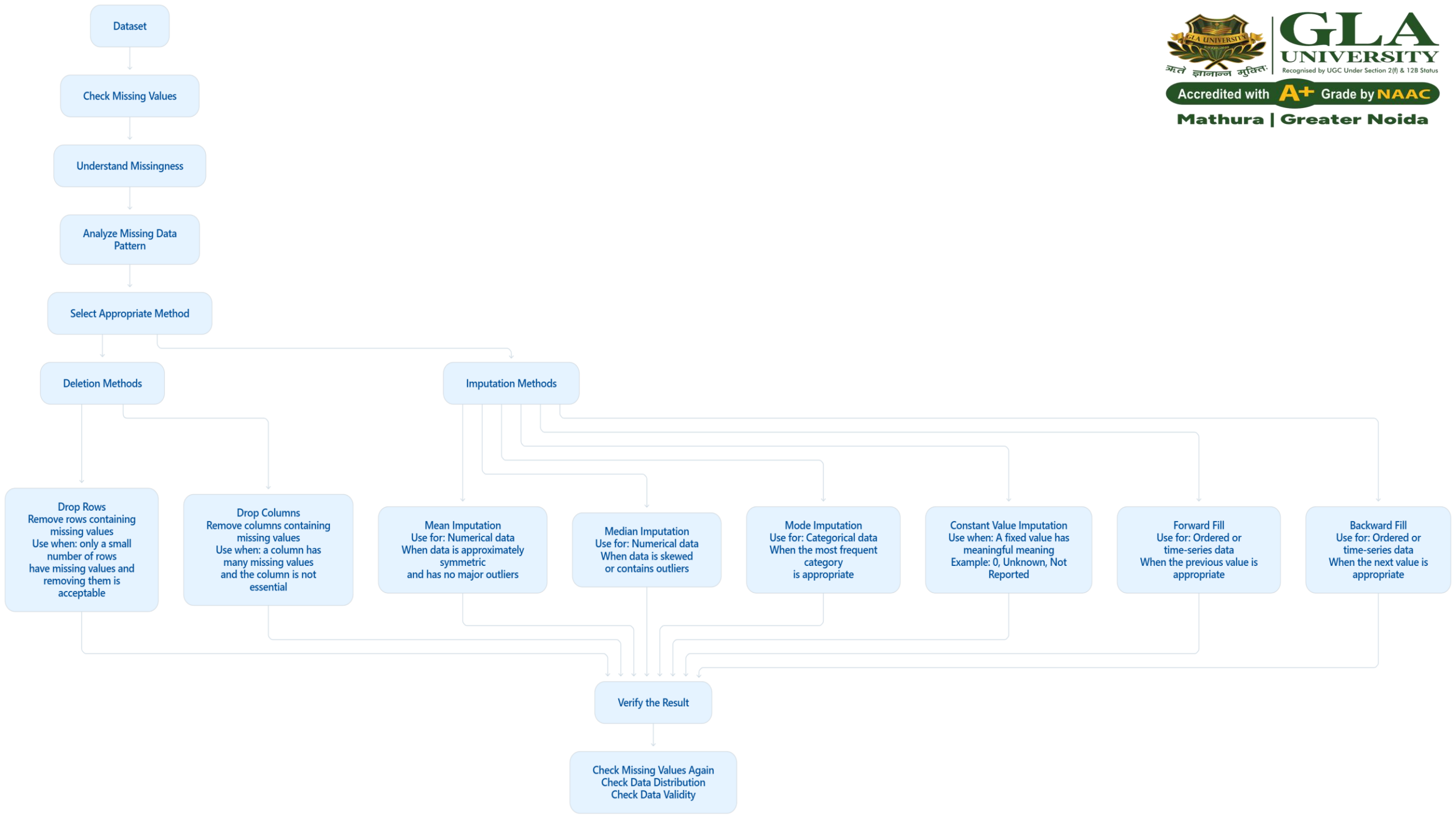
1. **Example**
    1. High earners may be less likely to disclose their income, leading to an underrepresentation of high-income data.
    2. Patients experiencing severe side effects might drop out of a clinical trial, causing the dataset to miss critical information.
2. **Why & How it Occurs?**
3. MNAR arises due to:
    1. Self-selection bias: Respondents intentionally omit sensitive or stigmatized information.
    2. Behavioral biases or decisions influenced by the missing data.

How to Solve?

MNAR is the most challenging type to handle because the missingness cannot be explained by observed data alone:

1. **Model-Based Approaches** :Use advanced statistical models to estimate and handle MNAR patterns. For example, mixed models or joint modeling approaches.
2. **Sensitivity Analysis** :Test various scenarios and assumptions about the missing data to understand its impact on results.
3. **Data Collection Improvements** :Attempt to reduce MNAR by redesigning data collection processes, such as anonymizing surveys or incentivizing responses.
4. **Pattern Analysis** :Investigate patterns in missingness to hypothesize and test potential reasons for MNAR.

Common Missing Data Handling Methods



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Checking Missing Values

Before applying any missing-value handling method, we first need to know:

- Which columns contain missing values?
- How many missing values are present?
- How much data is missing?

```
df.isnull() #  Detect missing values
```

	StudentName	StudentAge	End_term_marks	Department	Birth Year	DOB
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	False	False	False	False
3	False	False	False	False	False	False
4	False	False	False	False	False	False
6	False	False	False	False	False	False
7	False	False	False	False	False	False

```
df.isnull().sum() #  Detect missing values
```

	0
StudentName	0
StudentAge	0
End_term_marks	0
Department	0
Birth Year	0
DOB	0

dtype: int64

Deletion Method

Remove Missing Values : dropna(),Use dropna() when you decide that rows containing missing values should be removed.

```
#Remove rows
df.dropna()
```

	S_Name	S_Age	Marks
0	Amit	16.0	90.0
1	Reena	17.0	70.0
2	Shyam	18.0	55.0
4	abc	20.0	30.0
7	AAA	20.0	99.0
9	CCC	19.0	78.0

Removes rows containing missing values.

Delete Columns

It removes columns containing missing values.

```
#Remove columns
df.dropna(axis=1)
```

	S_Name
0	Amit
1	Reena
2	Shyam
3	Ravi
4	abc
5	xyz
6	wqr
7	AAA
8	BBB
9	CCC

axis=1 means columns

Mean Imputation

- Replace a missing numerical value with the mean (average) of that column.
- Used when ,Data is numerical.
- There are no major outliers.
- It is used when deletion is not suitable , and missing value percentage is more then 5%

```
df['Marks'] = df['Marks'].fillna(df['Marks'].mean())
```

```
df
```

	S_Name	S_Age	Marks
0	Amit	16.0	90.0
1	Reena	17.0	70.0
2	Shyam	18.0	55.0
3	Ravi	NaN	56.0
4	abc	20.0	30.0
5	xyz	21.0	66.5
6	wqr	23.0	54.0
7	AAA	20.0	99.0
8	BBB	NaN	66.5
9	CCC	19.0	78.0

All NaN values in the Marks column will be filled with the mean value.

Median Imputation

- Replace a missing numerical value with the median of the column.
- Use median imputation when : Data is numerical. Data contains outliers.
- The median better represents the typical value.

```
df['Marks'] = df['Marks'].fillna(df['Marks'].median())
```

```
df
```

	S_Name	S_Age	Marks
0	Amit	16.0	90.0
1	Reena	17.0	70.0
2	Shyam	18.0	55.0
3	Ravi	NaN	56.0
4	abc	20.0	30.0
5	xyz	21.0	63.0
6	wqr	23.0	54.0
7	AAA	20.0	99.0
8	BBB	NaN	63.0
9	CCC	19.0	78.0

Mode Imputation

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- Replace a missing value with the most frequently occurring value.
- Mainly used for categorical data.

Examples:

- Gender
- Department
- City
- Grade

```
df['Gender'] = df['Gender'].fillna(df['Gender'].mode()[0])
```

```
df
```

	S_Name	S_Age	Marks	Gender
0	Amit	16.0	90.0	Male
1	Reena	17.0	70.0	Female
2	Shyam	18.0	55.0	Male
3	Ravi	NaN	56.0	Male
4	abc	20.0	30.0	Female
5	xyz	21.0	NaN	Male
6	wqr	23.0	54.0	Female
7	AAA	20.0	99.0	Male
8	BBB	NaN	NaN	Male
9	CCC	19.0	78.0	Male

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Constant Value Imputation

- Replace the missing value with a fixed value.
- Useful when a meaningful fixed category/value is available

For numerical column

```
df['Marks'] = df['Marks'].fillna(0)
```

For Categorical

```
df['Department'] = df['Department'].fillna('Unknown')
```

Forward Fill : ffill()

- The missing value is replaced by the previously available value.
- Useful for ordered or time-series data, where the previous observation is relevant.

```
df['Marks'] = df['Marks'].ffill()
```

Backward Fill : bfill()

- The missing value is replaced by the next available value.
- Useful when the next observation is more appropriate for filling the missing value.

```
df['S_Age'] = df['S_Age'].bfill()
```

Duplicate Value Handling

- Duplicate data means the same record is repeated in a dataset.
- It may occur due to data entry, merging, or importing data.
- Duplicates can affect analysis and model performance.

Check Duplicate Values

- True : duplicate record
- False : not a duplicate

```
df.duplicated()
```

```
0    False
1    False
2    False
3    False
4    False
5    False
6    False
7    False
8     True
9    False
dtype: bool
```

Count duplicates:

```
df.duplicated().sum()
```

```
np.int64(1)
```

Remove Duplicate Rows

```
df = df.drop_duplicates()
```

```
df.drop_duplicates(inplace=True)
```